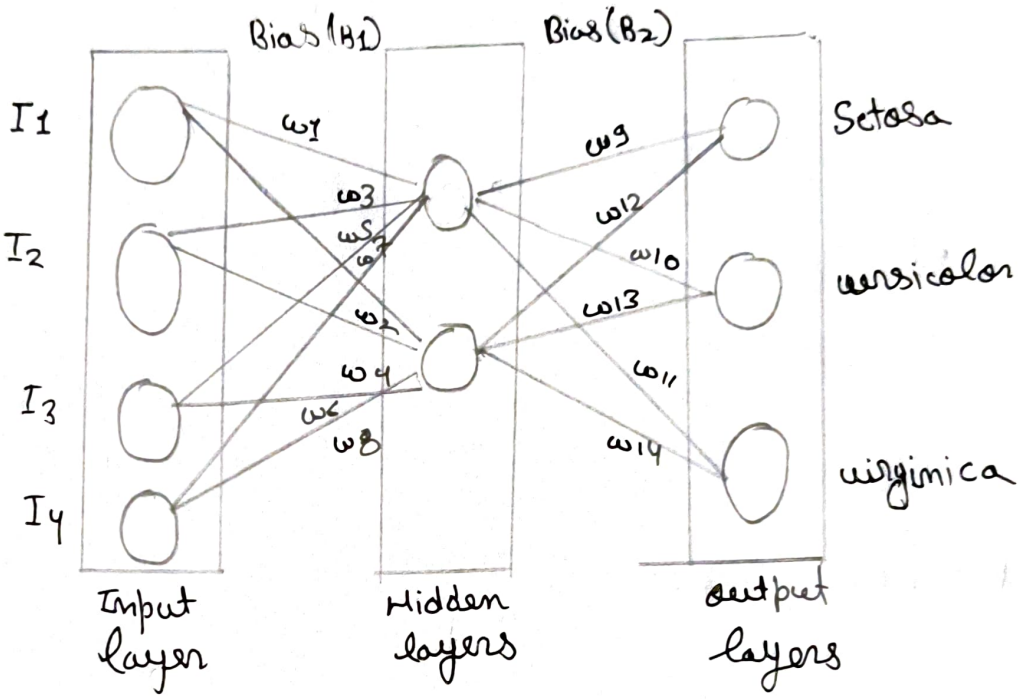
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Deep learning -

Assignment

=> Mathematical Calculation Single-Hidden layer neural network



1: Given Parameters

↳
- Input vector : $x = [5.1, 3.2, 1.4, 0.2]$
- Hidden ~~layer~~ : weights (H1) : $\omega_1 = 1$ $\omega_3 = 3$
  $\omega_5 = 5$ $\omega_7 = 7$
- Hidden weights (H2) : $\omega_2 = 2$ $\omega_4 = 4$
  $\omega_6 = 6$ $\omega_8 = 8$
- output weights (from H1) : $\omega_9 = 1.3$,
  $\omega_{10} = 0.5$ $\omega_{11} = 1.2$
- output weights (from H2) : $\omega_{12} = 0.5$
  $\omega_{13} = 0.2$ $\omega_{14} = 0.4$

- Biases : $b_1 = 2$ $b_2 = 0.1$

2. Hidden layer calculation

A) neuron 1 (H1):

↳ • linear combination (Z_1):

$$Z_1 = x_1 w_1 + x_2 w_3 + x_3 w_3 + x_4 w_4 + b_1$$

$$Z_1 = (5.1 \times 1) + (3.2 \times 3) + (1.4 \times 5) + (0.2 \times 7) + 2$$

$$Z_1 = 25.1$$

- Activation (σ_1) :

$$\sigma_1 = \frac{1}{1 + e^{-z_1}} = \frac{1}{1 + e^{-25.1}}$$

$$\sigma_1 = 1.0$$

B) neuron 2 (H2):

- linear combination (Z_2):

$$Z_2 = x_1 w_2 + x_2 w_4 + x_3 w_6 + x_4 w_8 + b_1$$

$$Z_2 = (5.1 \times 2) + (3.2 \times 4) + (1.6 \times 6) + (0.2 \times 8) + 2$$

$$Z_2 = 35.80$$

- Activation (σ_2):

$$\sigma_2 = \frac{1}{1 + e^{-z_2}} = \frac{1}{1 + e^{-35.0}}$$

$$\sigma_2 = 1.0$$

3. Output layer Calculation - (combining Both Neurons)

A) Setosa (z setosa):

$$z_{setosa} = (\sigma_1 \times w_9) + (\sigma_2 \times w_{12}) + b_2$$
$$= (1.0 \times 1.3) + (1.0 \times 0.5) + 0.1$$
$$= 1.3 + 0.5 + 0.1$$
$$z_{setosa} = 1.9$$

B) Versicolor (z versicolor):

$$z_{versicolor} = (\sigma_1 \times w_{10}) + (\sigma_2 + w_{13}) + b_2$$
$$= (1.0 \times 0.5) + (1.0 \times 0.2) + 0.1$$
$$= 0.5 + 0.2 + 0.1$$
$$z_{versicolor} = 0.8$$

(C) Virginica (z verginica):

$$z_{virginica} = (\sigma_1 \times w_{11}) + (\sigma_2 \times w_{14}) + b_2$$
$$= (1.0 \times 1.2) + (1.0 \times 0.4) + 0.1$$
$$= 1.2 + 0.4 + 0.1$$
$$z_{virginica} = 1.7$$

4. Final prediction -

- Output vector: [1.9, 0.8, 1.7]
- prediction: setosa [Highest Score 1.9]